

LAB# 9

Database Concepts

Database is a collection of related data and data is a collection of facts and figures that can be processed to produce information.

Database Management System or **DBMS** in short refers to the technology of storing and retrieving users data with utmost efficiency along with appropriate security measures.

Commands for DDL

CREATE

CREATE statements is used to define the database structure schema:

Syntax:

```
CREATE TABLE TABLE_NAME (COLUMN_NAME DATATYPES[,....]);
```

For example:

```
Create database university;  
Create table students;  
Create view for_students;
```

DROP

Drops commands remove tables and databases from RDBMS.

Syntax:

```
DROP TABLE ;
```

For example:

```
Drop object_type object_name;  
Drop database university;  
Drop table student;
```

ALTER

Alters command allows you to alter the structure of the database.

Syntax:

To add a new column in the table

```
ALTER TABLE table_name ADD column_name COLUMN-definition;
```

To modify an existing column in the table:

```
ALTER TABLE MODIFY(COLUMN DEFINITION....);
```

For example:

```
Alter table guru99 add subject varchar;
```

TRUNCATE:

This command used to delete all the rows from the table and free the space containing the table.

Syntax:

```
TRUNCATE TABLE table_name;
```

Example:

TRUNCATE table students;

Commands for DML

Here are some important DML commands:

- INSERT
- UPDATE
- DELETE

INSERT:

This is a statement that is a SQL query. This command is used to insert data into the row of a table.

Syntax:

INSERT INTO TABLE_NAME (col1, col2, col3,.... col N)
VALUES (value1, value2, value3, valueN);

For example:

INSERT INTO students (RollNo, FirstName, LastName) VALUES ('60', 'Tom', 'Erichsen');

UPDATE:

This command is used to update or modify the value of a column in the table.

Syntax:

UPDATE table_name SET [column_name1= value1,...column_nameN = valueN] [WHERE CONDITION]

For example:

UPDATE students
SET FirstName = 'Jhon', LastName=' Wick'
WHERE StudID = 3;

DELETE:

This command is used to remove one or more rows from a table.

Syntax:

DELETE FROM table_name [WHERE condition];

For example:

DELETE FROM students
WHERE FirstName = 'Jhon';